MULUNGUSHI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

ICT261: OBJECT-ORIENTED PROGRAMMING II — UNIT 4 LAB REPORT

STUDENT INFORMATION:

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- Program: Computer Science
- Date: November 2, 2026

ACTIVITY A: EXCEPTION PREDICTION & RISK ANALYSIS

When reading external text files and parsing student grade records line by line, five core exception scenarios can occur:

1. FileNotFoundException:
   Triggered if the specified input file ("marks.txt") is missing from the directory or if the file name/path is misspelled.

2. IOException:
   Occurs during low-level input/output operations, such as sudden file stream interruptions, unreadable storage media, or restricted access permissions.

3. NumberFormatException:
   Raised when Integer.parseInt() attempts to parse non-numeric input string data (e.g., "ABC" or decimal values) into integer grades.

4. IllegalArgumentException:

Thrown during input validation checks if a record does not contain exactly two comma-separated values, or if a parsed mark falls outside the range (0–100).

5. DuplicateStudentException:

A custom unchecked runtime exception thrown when a student ID is encountered more than once within the dataset

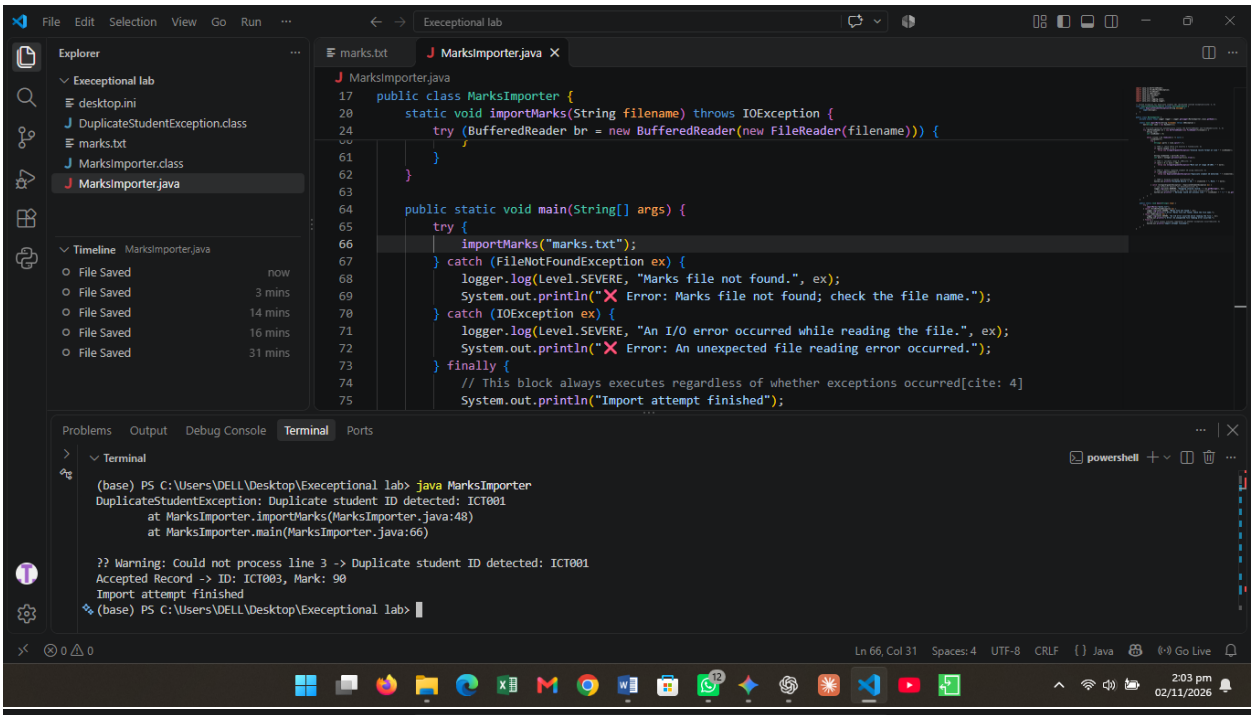
ACTIVITY D: ENGINEER'S REVIEW & EXCEPTION STRATEGY

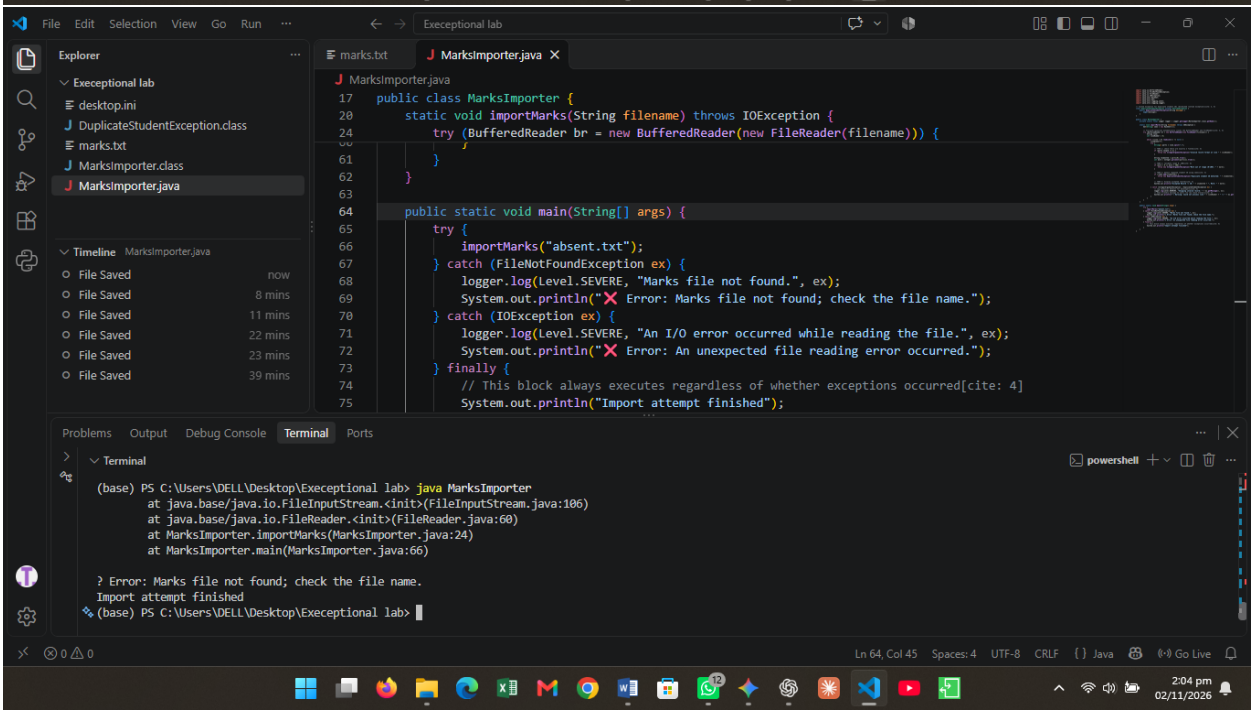
1. Why skip invalid records rather than aborting the entire import process?

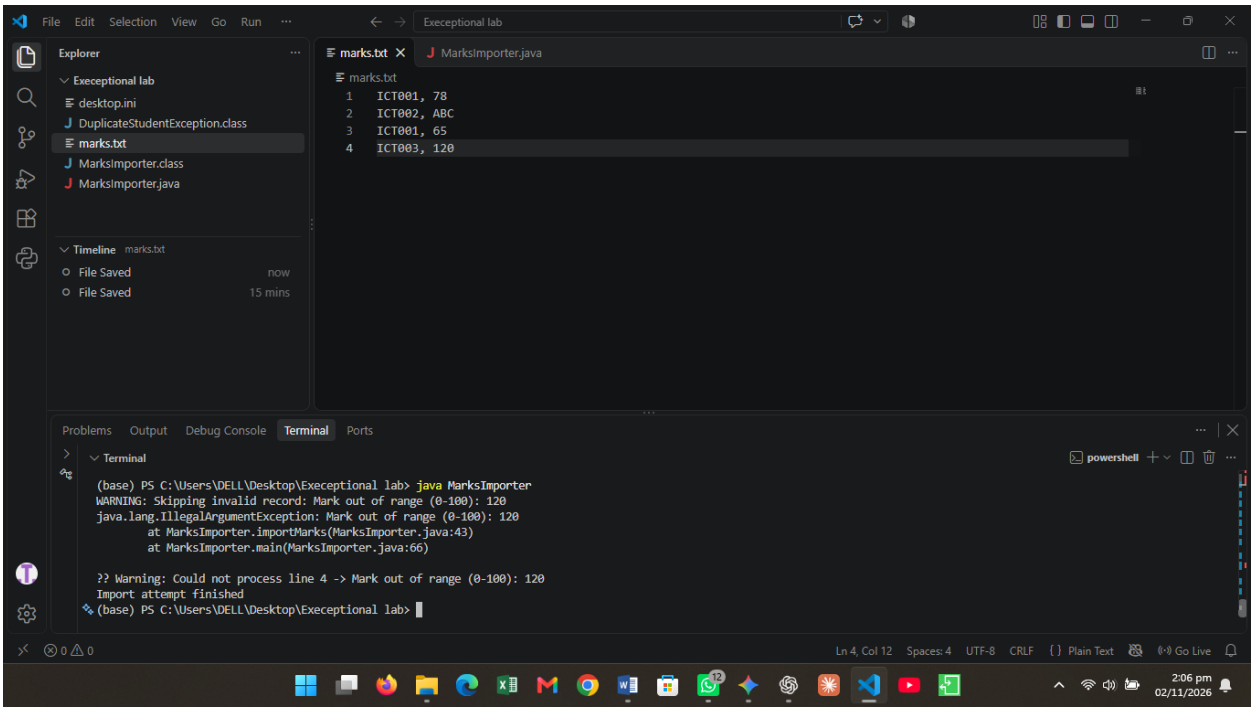
In enterprise applications and academic batch processing, fault tolerance is critical. Halting an entire import operation because of a single bad line or typo leads to unnecessary system failure and administrative inefficiency. By catching exceptions locally per record line, logging technical details via the Java Logging API, and showing user-friendly warning messages, the program safely skips corrupt entries while processing all valid data cleanly.

2. Summary of Executed Test Scenarios (T1–T5):

- T1 (Valid Input): Successfully processed and displayed valid student records.
- T2 (Invalid Format/Type): Caught non-integer marks and logged line warnings.
- T3 (Out-of-Range Marks): Caught marks outside 0–100 and skipped corrupt rows.
- T4 (Duplicate Student ID): Caught duplicate IDs using a HashSet and threw DuplicateStudentException.
- T5 (Missing File Handling): Caught FileNotFoundException for "absent.txt", displayed error message, and executed the 'finally' cleanup block.


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
marks.txt
MarksImporter.class
MarksImporter.java
Timeline MarksImporter.java
File Saved now
File Saved 3 mins
File Saved 14 mins
File Saved 16 mins
File Saved 31 mins
marks.txt
MarksImporter.java
MarksImporter.java
17 public class MarksImporter {
20 static void importMarks(String filename) throws IOException {
24 try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
61 }
62 }
63
64 public static void main(String[] args) {
65 try {
66 importMarks("marks.txt");
67 } catch (FileNotFoundException ex) {
68 logger.log(Level.SEVERE, "Marks file not found.", ex);
69 System.out.println("❌ Error: Marks file not found; check the file name.");
70 } catch (IOException ex) {
71 logger.log(Level.SEVERE, "An I/O error occurred while reading the file.", ex);
72 System.out.println("❌ Error: An unexpected file reading error occurred.");
73 } finally {
74 // This block always executes regardless of whether exceptions occurred[cite: 4]
75 System.out.println("Import attempt finished");
Problems Output Debug Console Terminal Ports
Terminal
powershell
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
DuplicateStudentException: Duplicate student ID detected: ICT001
at MarksImporter.importMarks(MarksImporter.java:48)
at MarksImporter.main(MarksImporter.java:66)
?? Warning: Could not process line 3 -> Duplicate student ID detected: ICT001
Accepted Record -> ID: ICT003, Mark: 90
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
Ln 66, Col 31 Spaces: 4 UTF-8 CRLF Java Go Live
2:03 pm
02/11/2026


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
marks.txt
MarksImporter.class
MarksImporter.java
Timeline MarksImporter.java
File Saved now
File Saved 8 mins
File Saved 11 mins
File Saved 22 mins
File Saved 23 mins
File Saved 39 mins
marks.txt
MarksImporter.java
MarksImporter.java
17 public class MarksImporter {
20 static void importMarks(String filename) throws IOException {
24 try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
61 }
62 }
63
64 public static void main(String[] args) {
65 try {
66 importMarks("absent.txt");
67 } catch (FileNotFoundException ex) {
68 logger.log(Level.SEVERE, "Marks file not found.", ex);
69 System.out.println("❌ Error: Marks file not found; check the file name.");
70 } catch (IOException ex) {
71 logger.log(Level.SEVERE, "An I/O error occurred while reading the file.", ex);
72 System.out.println("❌ Error: An unexpected file reading error occurred.");
73 } finally {
74 // This block always executes regardless of whether exceptions occurred[cite: 4]
75 System.out.println("Import attempt finished");
Problems Output Debug Console Terminal Ports
Terminal
powershell
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
at java.base/java.io.FileInputStream.<init>(FileInputStream.java:106)
at java.base/java.io.FileReader.<init>(FileReader.java:60)
at MarksImporter.importMarks(MarksImporter.java:24)
at MarksImporter.main(MarksImporter.java:66)
? Error: Marks file not found; check the file name.
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
Ln 64, Col 45 Spaces: 4 UTF-8 CRLF Java Go Live
2:04 pm
02/11/2026


File Edit Selection View Go Run
Exceptional lab
Explorer
Exceptional lab
desktop.ini
DuplicateStudentException.class
marks.txt
MarksImporter.class
MarksImporter.java
Timeline marks.txt
File Saved now
File Saved 15 mins
marks.txt
MarksImporter.java
marks.txt
1 ICT001, 78
2 ICT002, ABC
3 ICT001, 65
4 ICT003, 120
Problems Output Debug Console Terminal Ports
Terminal
powershell
(base) PS C:\Users\DELL\Desktop\Exceptional lab> java MarksImporter
WARNING: Skipping invalid record: Mark out of range (0-100): 120
java.lang.IllegalArgumentException: Mark out of range (0-100): 120
at MarksImporter.importMarks(MarksImporter.java:43)
at MarksImporter.main(MarksImporter.java:66)
?? Warning: Could not process line 4 -> Mark out of range (0-100): 120
Import attempt finished
(base) PS C:\Users\DELL\Desktop\Exceptional lab>
Ln 4, Col 12 Spaces: 4 UTF-8 CRLF Plain Text Go Live
2:06 pm
02/11/2026